1️⃣ Manipulate numbers, graphical and statistical data (Item 7)

Relevance: ★★★★★ (Most critical)

Description:

> Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations.

Manifestation in the problem:

Students must manipulate numerical information including:

- Distances (4 km, 3 km)
- Time values (9 min, 6 min)
- Conversion of minutes to hours
- Calculation of total distance and total time

This requires performing operations such as:

- Addition
- Division
- Ratio calculation

Therefore, **numerical manipulation is the core competency** in this task.

2️⃣ Use a multi-step procedure leading to a mathematical solution (Item 12)

Relevance: ★★★★★

Description:

> Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation.

This problem requires several steps:

1. Identify the two segments of the trip
2. Add the distances to find the total distance
3. Add the times to find the total time
4. Convert minutes into hours
5. Apply the average speed formula
6. Compute the final result

Thus, solving the problem involves a **structured multi-step reasoning process**.

3️⃣ Understand and utilise constructs based on definitions and rules (Item 5)

Relevance: ★★★★☆

Description:

> Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms.

Students must understand the **definition of average speed**:

$$Average\ speed = \frac{total\ distance}{total\ time}$$

This requires recognizing that the **average speed of the entire trip must use total distance and total time**, rather than averaging the two segment speeds.

This conceptual understanding is essential for correct modeling.

4️⃣ Perform a simple calculation (Item 1)

Relevance: ★★★★☆

Description:

> Perform a simple calculation.

The task involves several basic calculations:

- $4 + 3$
- $9 + 6$
- $15 \div 60$
- $7 \div 0.25$

Although each step is simple, they must be combined correctly to obtain the final answer.

5️⃣ Make sense of and use representations (Item 10)

Relevance: ★★★☆☆

Description:

> Make sense of, relate and use a variety of representations when interacting with a problem.

Students convert the **verbal description** into mathematical expressions:

Text information → numerical data → formula application.

This transformation between representations supports the problem-solving process.

6️⃣ Use an understanding of the context to guide solving (Item 13)

Relevance: ★★★☆☆

Description:

> Use an understanding of the context to guide or expedite the mathematical solving process.

Students must interpret the real-world context:

- Bicycle travel
- Two different routes
- Distance and time relationships

Understanding the trip structure helps students determine that **the whole-trip average speed must be calculated using totals**.

1️⃣ Manipulate numbers, graphical and statistical data（第7项）

相关性：★★★★★（最核心）

能力描述：

> Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations.

在本题中的体现：

学生需要处理和运算多组数值信息，包括：

- 距离（4 km、3 km）
- 时间（9 分钟、6 分钟）
- 单位转换（分钟 → 小时）
- 总距离与总时间的计算

涉及的运算包括：

- 加法
- 除法
- 比例计算

因此，**数值操作能力是本题最核心的能力。**

2️⃣ Use a multi-step procedure leading to a mathematical solution（第12项）

相关性：★★★★★

能力描述：

> Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation.

本题需要多个步骤才能完成求解：

1. 识别行程的两个阶段
2. 计算总距离
3. 计算总时间
4. 将分钟转换为小时
5. 使用平均速度公式
6. 计算最终结果

因此，该题属于**多步骤推理求解问题。**

3️⃣ Understand and utilise constructs based on definitions and rules（第5项）

相关性：★★★★☆

能力描述：

> Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms.

学生必须理解 **平均速度的定义**：

$$\text{平均速度} = \frac{\text{总距离}}{\text{总时间}}$$

关键点在于：

平均速度必须用 **总距离 ÷ 总时间**，
而不是简单地对两段速度进行平均。

这种概念理解对于正确建模非常重要。

4️⃣ Perform a simple calculation（第1项）

相关性：★★★★☆

能力描述：

> Perform a simple calculation.

题目涉及多次基础计算：

- $4 + 3$
- $9 + 6$
- $15 \div 60$
- $7 \div 0.25$

虽然每一步运算都比较简单，但必须正确组合才能得到最终答案。

5️⃣ Make sense of and use representations（第10项）

相关性：★★★☆☆

能力描述：

> Make sense of, relate and use a variety of representations when interacting with a problem.

学生需要将：

文字情境描述 → 数学表达式

即：

文本信息 → 数值信息 → 数学公式。

这种表征转换支持问题求解过程。

6️⃣ Use an understanding of the context to guide solving（第13项）

相关性：★★★☆☆

能力描述：

> Use an understanding of the context to guide or expedite the mathematical solving process.

学生需要理解现实情境：

- 骑行行程
- 不同路线
- 距离与时间关系

理解整个行程结构有助于学生认识到：

整个往返的平均速度需要使用总距离和总时间计算。